ZHISHAO NI

Master's Student | Autonomous Driving Motion Planning and Safe Decision Control

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PROFILE

Graduate researcher in autonomous driving, with a focus on motion planning, reinforcement learning, and safe and efficient decision control in complex interactive traffic environments. Experienced in integrating model-based expert knowledge with data-driven learning methods.

EDUCATION

Master of Engineering in Mechanical Engineering (expected) **Sep 2024 - Jun 2027**

East China Jiaotong University, Nanchang, China

Bachelor of Engineering in Vehicle Engineering **Sep 2019 - Jun 2023**

Anhui Science and Technology University, Anhui, China

RESEARCH INTERESTS

Autonomous driving motion planning; safe and intelligent decision control; reinforcement learning; model predictive control; safety-efficiency co-optimization in complex interactive traffic environments.

MASTER'S RESEARCH PROJECT

Physics-informed reinforcement learning for autonomous-driving merging decision and control

Developing a ramp-merging decision and control framework that combines model predictive control expert priors with residual reinforcement learning. The work uses risk-adaptive execution, task-safety value decoupling, and curriculum learning to improve training stability, sample efficiency, and safe and efficient merging performance.

JOURNAL PUBLICATIONS

1. D. Zeng, **Z. Ni**, Z. Li*, X. Zhang, M. Liu, Y. Hu, C. Sun, and B. Leng, "Physics-informed residual reinforcement learning via expert prior knowledge for safe and efficient autonomous merging," *Neurocomputing*, vol. 701, art. 134631, 2026. doi:10.1016/j.neucom.2026.134631 (JCR Q1; second author)

2. D. Zeng*, **Z. Ni**, P. Zhang, Y. Hu, L. Gao, T. Ai, and C. Liang, "A Unified Optimal Plan-control Method Addressing Arbitrary Lots for Parking Autonomous Vehicle," *International Journal of Automotive Technology*, 2026. doi:10.1007/s12239-026-00463-5 (JCR Q3; second author)

** Corresponding author*

CONFERENCE PAPER

1. D. Zeng*, **Z. Ni**, J. Lu, et al., "Speed-Scheduled Predictive Yaw Stabilization for a Four-Axle Electric Wheel-Driven Vehicle," *The 3rd International Conference on Intelligent Systems and Robotics*. (EI indexed; second author)

ZHISHAO NI

Research Output and Qualifications

PATENTS

1. Granted. "An MPC-guided reinforcement learning method for autonomous-driving ramp merging." Inventors: Dequan Zeng, **Zhishao Ni**, Yiming Hu, and Jinwen Yang. East China Jiaotong University, 2026.

2. Published application. "A phase-adaptive reinforcement learning method for autonomous-driving ramp merging." Inventors: Dequan Zeng, **Zhishao Ni**, Junyu Zhou, Jun Lu, Yiming Hu, and Jinwen Yang. East China Jiaotong University, 2026.

3. Under examination. "An autonomous-vehicle merging method based on a candidate-gap lane-changing potential function." Inventors: Dequan Zeng, **Zhishao Ni**, Junyu Zhou, Yiming Hu, and Jinwen Yang. East China Jiaotong University, 2026.

HONORS AND AWARDS

2026	First Prize, Jiangxi Provincial Graduate Mathematical Contest in Modeling
2025	Grand Prize, Jiangxi Provincial Graduate Mathematical Contest in Modeling
2025	Jiangxi Provincial Graduate Academic Scholarship
2024	East China Jiaotong University Graduate Academic Scholarship

TECHNICAL SKILLS

- **Programming and scientific computing:** Python, MATLAB
- **Engineering and design:** AutoCAD, SolidWorks
- **Productivity:** Microsoft Office
- **Research capabilities:** reinforcement learning, model predictive control, autonomous-driving decision and control, mathematical modeling

LANGUAGES

Chinese: Native | English: CET-4 and CET-6

PROFESSIONAL STRENGTHS

Systematic academic training with strong research, innovation, and engineering-practice capabilities. Effective communicator and collaborative team member with experience coordinating technical work across research projects.